

DESIGN AND CONTROL OF AZIMUTH THRUSTERS FOR AUTONOMOUS
SURFACE VEHICLES

By

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To my parents and friends for their invaluable support throughout my time at UF

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LIST OF ABBREVIATIONS

AUV	Autonomous Underwater Vehicle
ASV	Autonomous Surface Vehicle
UAV	Unmanned Aerial Vehicle
ESC	Electronic Speed Controller

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The PropaGator 3 autonomous surface vehicle (ASV) will compete in the RobotX 2026 challenge. Part of this challenge required the ASV to maneuver around obstacles. The Blue Boat platform, which PropaGator 3 is currently built on only has two fixed thrusters, which create a nonholonomic constraint. Azimuth thrusters provide a solution to this issue. This thesis details the design and fabrication of a prototype azimuth thruster. Additionally, this thesis introduces a solution using cost functions to optimally allocate thrust vectors between the fixed thrusters and the azimuth thrusters.

CHAPTER 1 INTRODUCTION

Autonomous systems are becoming increasingly utilized for both commercial and military applications. They allow for long duration tasks to be completed without putting humans in dull, dirty, or dangerous situations. In maritime environments, autonomous systems are used to monitor environmental conditions, survey underwater objects, and patrol waters around military installations.

The RobotX Challenge

To prepare the next generation of engineers to develop these increasingly useful systems, RoboNation, an American nonprofit, has created a series of international collegiate competitions. The most complex of these competitions is RobotX, where an autonomous underwater vehicle (AUV), autonomous surface vehicle (ASV), and an unmanned aerial vehicle (UAV), must autonomously collaborate to accomplish a series of challenges. The RobotX challenge is held bi-annually, with the next competition at the time of writing in Singapore in November 2026.

The University of Florida's Machine Intelligence Laboratory (MIL) is developing SubjuGator 9, PropaGator 3, and InvestiGator as its AUV, ASV, and UAV entries, respectively. While SubjuGator 9 is an existing vehicle prior to the 2026 competition, PropaGator 3 is being built from the ground up for RobotX 2026. This allows a large amount of flexibility and control over the design of the vehicle to tailor it to the challenges in the competition.

The focus of this paper is the ASV. There are two maneuverability challenges (fig. 1-1) that the ASV must complete at RobotX. The first is a channel navigation task

where an array of buoys are set up in the water. Each buoy has a light on top only visible from the air. The UAV must use its camera to observe the color of each light, then relay this information to the ASV so it can navigate through the channel. Throughout the duration of the challenge, the buoys lights may change, requiring the ASV to plot a new path, adapting to the changing lights.

The second challenge is a docking task. After receiving a color code, the ASV must navigate to the corresponding dock, then navigate inside the dock. While docked, the ASV must use a launcher to shoot one liter of water at a target.

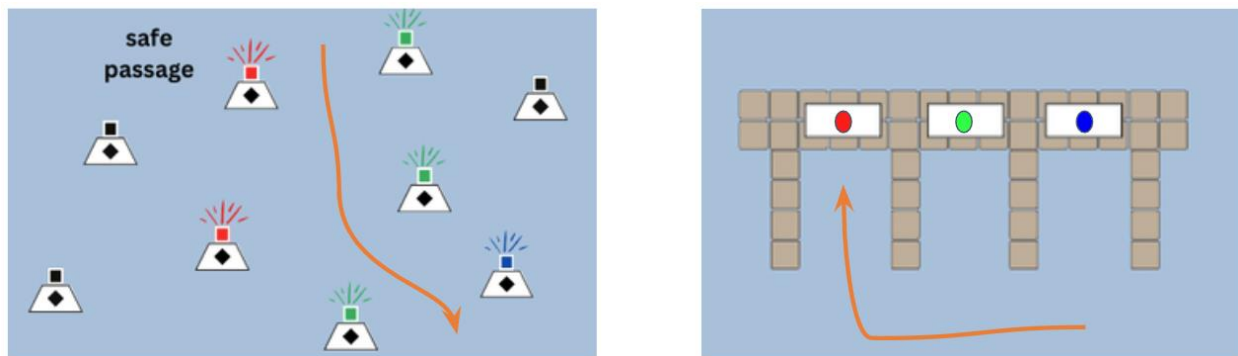


Fig. 1-1 ASV Challenges at RobotX 2026. Navigation channel (left) and docking (right)

PropaGator 3

PropaGator 3 is the Machine Intelligence Lab's entry to the RobotX 2026 competition. Due to limited time to prepare for the competition, the lab chose to purchase a base ASV platform, then modify its electrical, sensing, and propulsion systems. The base ASV chosen was the Blue Boat platform (fig. 1-2) sold by Blue Robotics. The Blue Boat is a catamaran ASV with the electrical systems contained inside the pontoons. There are mounting points for additional hardware both on top of and between the pontoons.



Fig. 1-2. Blue Boat Platform

The Blue Boat comes fitted with two fixed thrusters driven by Blue Robotics M200 motors. By having these two fixed thrusters, the vehicle's movement space is restricted via a nonholonomic constraint (fig. 1-3).

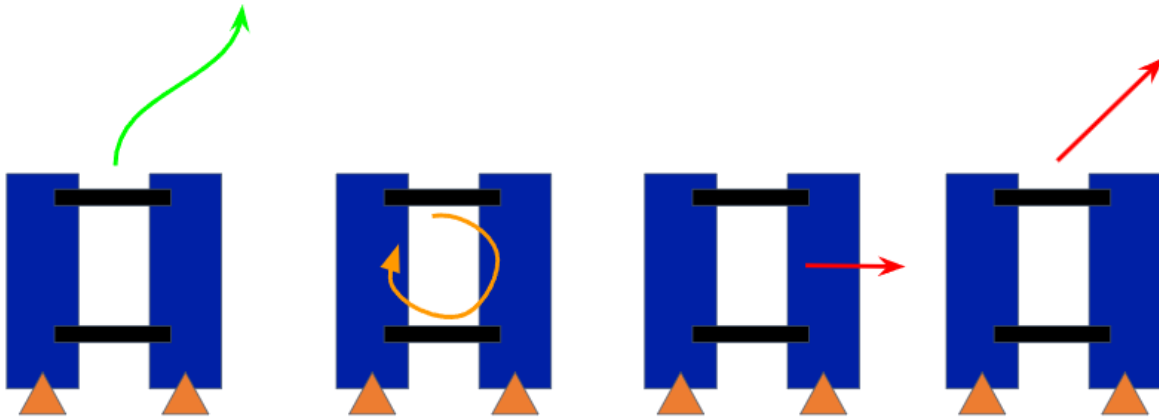


Fig. 1-3. Movement capabilities of the Blue Boat with fixed thrusters

The movement capabilities of the Blue Boat in a world frame can be represented mathematically by Equation 1, where v is the linear velocity as a result of averaging each thruster's velocity (v_1 and v_2), $\dot{\theta}$ is the angular velocity as a result of the thrusters moving at different speeds. $\dot{x}$ and $\dot{y}$ are the translational velocity in the world frame,

and θ is the heading of the ASV relative to the world. Figure 1-4 shows this setup visually.

$$\vec{v}_R = \begin{bmatrix} v \\ 0 \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \quad (1)$$

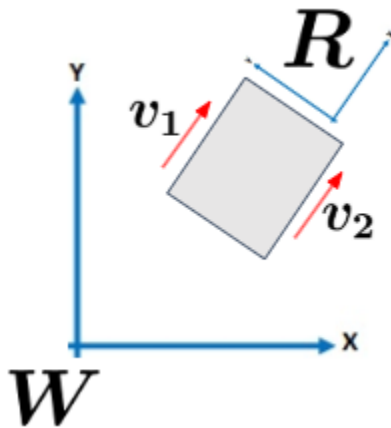


Fig. 1-4. Setup of robot R in world frame W

The nonholonomic constraint is given by (2) and represents the robot's incapability to instantaneously translate in the y (sideways) direction.

$$v_{R,y} = -\dot{x} \sin(\theta) + \dot{y} \cos(\theta) = 0 \quad (2)$$

The goal of this project is to eliminate this nonholonomic constraint. There are a number of options for generating a sideways component of velocity. Vectored (angled) thrusters could be added to the ASV to allow a component of thrust to act in the y direction. This approach would be difficult to accomplish in the available timeframe because it would require a significant mechanical redesign of much of the Blue Boat

platform. In addition, vectored thrusters have significantly reduced effectiveness due to a reduced surface area relative to orthogonal thrusters.

Alternatively, an additional thruster perpendicular to the fixed thruster could be added. While this thruster would be effective while translating sideways, it would introduce additional drag that the fixed thrusters would have to work against. The sideways thruster would also not be able to contribute to forwards or backwards motion, which make up the majority of necessary thrust cases. A third option is to use azimuth thrusters, which were used on PropaGator 2 [1]. Azimuth thrusters allow the thruster to be rotated to any direction, reducing drag and maximizing the thruster's effectiveness.

Azimuth Thrusters

Azimuth thrusters (fig. 1-5) are a form of thruster that can rotate about a central axis.



Fig. 1-5. Example of an azimuth thruster

By adding this additional component of thrust, the nonholonomic constraint previously discussed can be broken, as shown in (3), where v_a is the net thrust from the azimuth thruster(s) and ϕ is the net azimuth angle. ω_a is the resultant rotation caused by the azimuth thruster's line of action not intersecting the center of mass of the vehicle.

$$\vec{v}_R = \begin{bmatrix} v \\ 0 \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} v_a \cos(\phi) \\ v_a \sin(\phi) \\ \omega_a \end{bmatrix} \quad (3)$$

For this thesis, the system will use two azimuth thrusters, mounted along the centerline at the bow and stern of the vehicle.

Previous Design

Azimuth thrusters were previously designed for PropaGator 2 (fig. 1-6). These thrusters were highly effective at the RoboBoat competition and gave PropaGator 2 an unparalleled level of maneuverability. The thrusters used a majority of custom design parts, including a machined Kort nozzle propeller design, as well as belt driven azimuth rotation and propulsion.



Fig. 1-6. PropaGator 2 azimuth thruster design

Despite the technical success of the thruster, it suffered from multiple operational limitations. The thruster was mounted inside the hull of the vehicle and sealed at the interface. While this was effective at reducing drag and minimizing the wetted surface area of the thruster, it meant that if the thruster were removed, the vehicle would no longer be waterproof. Additionally, being mounted inside the boat made the thruster difficult to service, which could take hours. The thruster also has a large footprint (fig. 1-7), and while not an issue for the larger hull of PropaGator 2, PropaGator 3 has limited hull space available for a large internal mechanism.



Fig. 1-7. Side view (left) and top view (right) of the PropaGator 2 azimuth thruster

These limitations in PropaGator 2's azimuth thrusters led to the creation of design goals for a new generation of azimuth thrusters. The design goal is that the new thrusters enable omnidirectional movement. Without this capability, there is no reason to add new thrusters. To minimize downtime during the limited period before the RobotX competition, the thrusters must also be removable from the vehicle without disabling it. The final design goal is that the thrusters must be able to integrate with the existing mechanical and electrical infrastructure on PropaGator 3.

CHAPTER 2

SYSTEM DESIGN

This chapter details the design process and prototype creation for the new azimuth thrusters. There were three major iterations of the mechanical design (fig. 2-1).

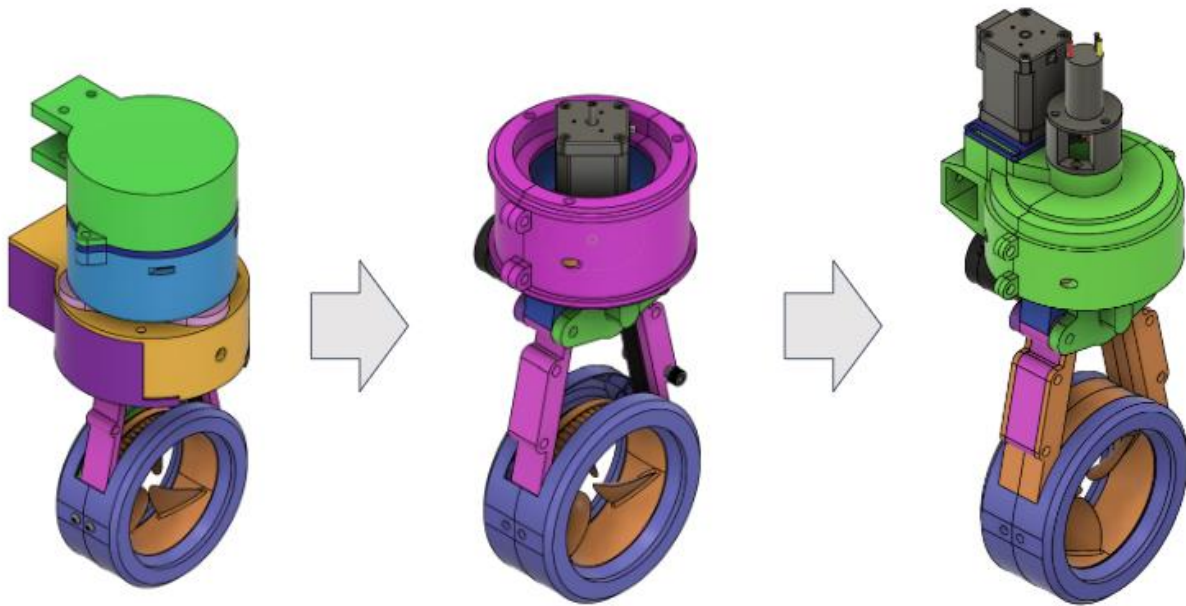


Fig. 2-1. Mechanical design iterations for the azimuth thruster, from left to right

The first iteration helped to define the size and form of the thruster, as well as how components would interact with each other. The second iteration introduced the selected actuators – a Blue Robotics M200 motor to drive the propeller’s rotation, and a stepper motor to drive the azimuth rotation. The third (and at the time of writing final) iteration of the thruster expanded on the second by providing mounting points, protection for spinning components, and other hardware.

Design Overview

The final design uses 11 custom parts, all of which were able to be 3D printed for the prototype. The system can be split into two parts: the lower propeller assembly and the upper azimuth assembly.

The propeller assembly (fig. 2-2) is responsible for housing the propeller and rigidly connecting it via a belt to the azimuth assembly. The bearing trays (blue) each hold three small roller bearings. These bearings are held in place by the bearing plate (green). The bearing plate has a small interface with the stationary inner ring of each bearing and is held in place by six bolts. The bearings are the only point of contact with the propeller, and interface with it via an angled face on the propeller. The propeller ring (pink) is inserted around the propeller and attached to the bearing tray via two bolts on each side. The assembly is then mirrored.

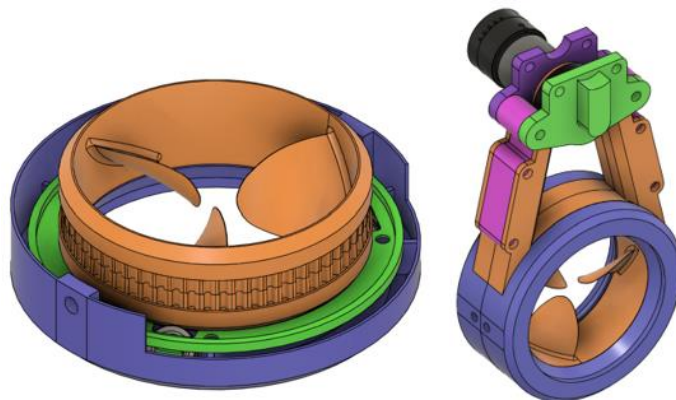


Fig. 2-2. Bearing tray assembly (left) and complete propeller assembly (right)

To connect to the M200 motor, two plates are mounted to the top of the bearing ring, with one holding the motor and another holding a bearing to support the motor's shaft. A custom HTD gear interfaces with the motor's D-shaft to drive the belt.

The azimuth assembly (fig. 2-3) is responsible for rotating the azimuth thruster to a desired orientation and then rigidly holding it there. The azimuth base (blue) connects to the propeller ring from the propeller assembly by two bolts. There is a cable path for the M200 motor's cable to be routed through the assembly. On top of the assembly is a slip ring, allowing continuous and unbounded rotation without damaging the motor's cable. The azimuth rotation is driven by a stepper motor. The stepper motor directly interfaces with a 26-tooth gear, which then interfaces with the azimuth gear (pink), a 100-tooth custom gear. To allow easy maintenance, the azimuth assembly can be slid out of its housing after the clamp (orange) is loosened. The bolts from the clamp are externally accessible from the holes in the housing (green). The bolts always align with the external holes because of a small groove on the clamp.

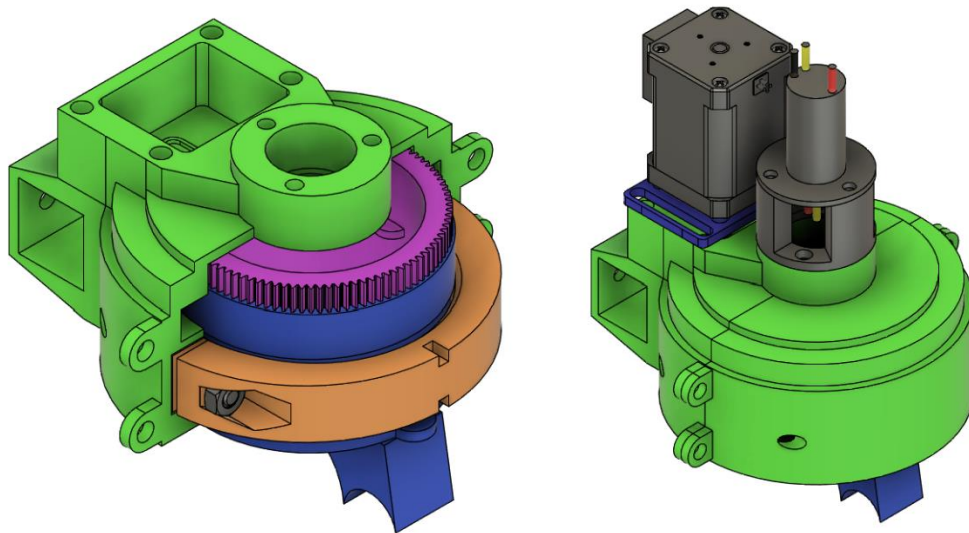


Fig. 2-3. Azimuth assembly with exposed internals (left) and complete (right)

The stepper motor also uses an absolute encoder. While stepper motors are capable of internally tracking their position (in number of steps which can be converted

to position), the tracking is reset when the motor loses power. Without an absolute encoder, the thruster would need to be set to a predefined “zero” position so it could be accurately controller. By adding an absolute encoder, this limitation is avoided and the thruster’s absolute position can be determined at startup.

A larger image of the full assembly is shown in figure 2-4.

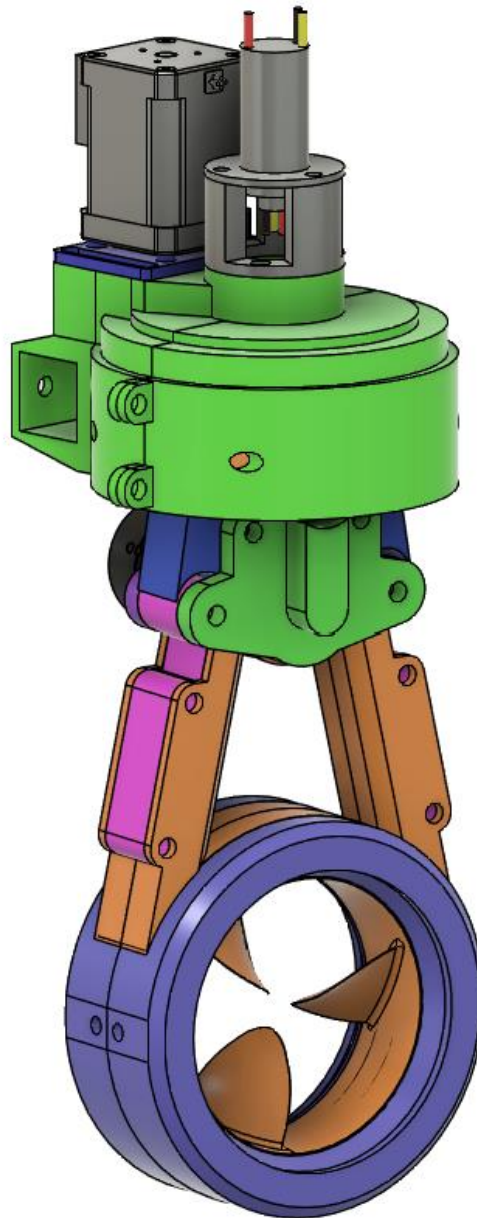


Fig. 2-4. Full CAD assembly of the azimuth thruster

Prototype Assembly

A full prototype (fig. 2-5) of the azimuth thruster was created via 3D printing using a Bambu X1C 3D printer. All parts were printed in PETG filament. One error in the previous design discovered during prototype assembly was that the M200 motor's shaft does not spin, but rather the entire rotor and shaft upper assembly spins. This issue was resolved by offsetting the M200 motor stator mounting using two spacers and a spare motor mounting plate.



Fig. 2-5. Azimuth thruster prototype

Both motion elements were tested with the prototype. The propeller was driven by connecting a Blue Robotics basic Electronic Speed Controller (ESC) to the M200 motor and generating a PWM signal using a stock servo tester. During testing, the motor was spun at 1700 – 1800 RPM and successfully drove the propeller.

The azimuth rotation was also tested by connecting the stepper motor to a a4988 stepper motor driver and generating a command signal with an Arduino Nano. The azimuth rotation was able to repeatably spin to desired rotation angles.

The thruster was successfully able to accomplish both movement goals in a lab environment. Further testing would be required to determine the effectiveness of the thruster in water when mounted to PropaGator 3, including how water affects the belt interfaces and how much thrust the azimuth thruster produces.

Wear and Fatigue

As with all mechanical systems, the thruster will undergo wear as a result of friction between moving parts. The two parts that see the highest wear are the small bearings that the propeller rests on and the belt. A simple fatigue analysis is conducted for both of them.

The industry standard for bearing lifetime is given by an L10 rating, which gives the lifetime of the bearing in millions of revolutions. The L10 rating represents the runtime when 90% of bearings would still be operational, so 10% of bearings will fail before this. The L10 rating is calculated with (4), where C is the dynamic loading capacity of the bearing, P is the equivalent load on the bearing, and p is a scaling factor determined by the type of bearing. For ball bearings, p is 3.

$$L_{10} = \left(\frac{C}{P}\right)^p \quad (4)$$

440C stainless steel corrosion resistant bearings were selected for the prototype because of their high dynamic loading capacity (260) and sealed faces. With these bearings and an approximate maximum load of 30 lbs per bearing, the L10 rating was

calculated to be 650. This can be converted to thruster operational life L in hours with (5), with the M200 motor RPM assumed to be a constant maximum 2700. For the prototype, the diameters of the gear and bearing are given by $D_{gear} = 1.65 \text{ in}$ and $D_{bearing} = 0.625 \text{ in}$, respectively. This lifetime is likely longer than the operational lifetime of PropaGator 3, and assumes the thruster is being driven at its maximum thrust, so the actual lifetime may be longer.

$$L = \frac{L_{10}}{\frac{D_{gear}}{D_{bearing}} \cdot RPM_{motor}} \cdot \frac{1 \text{ hr}}{60 \text{ min}} \approx 1500 \text{ hr} \quad (5)$$

For the belt that drives the propeller, the HTD belt standard is for the belt to undergo between 5,000,000 and 15,000,000 cycles. At the average value of 10,000,000 belt cycles. The expected lifetime L of the belt in hours is given by (6), where C_{gear} is the circumference of the gear driven by the motor and L_{belt} is the length of the belt.

$$L = \frac{10,000,000}{\frac{C_{gear}}{L_{belt}} \cdot RPM_{motor}} \cdot \frac{1 \text{ hr}}{60 \text{ min}} \approx 200 \text{ hr} \quad (6)$$

Electrical System

While not the focus of this thesis project, the electrical system was also given consideration. One of the driving forces behind using a Blue Robotics M200 motor was that they were already used on the Blue Boat. By using the same motors, the same ESCs and power delivery could be used. Additionally, from a logistics view, when

travelling around the world for competitions, it is easier to have a standardized electronics system. Because of this, the only new component needed is the motor driver for the stepper motor. An electrical block diagram is shown in figure 2-6. The NVIDIA Jetson Orin used as the main computer on PropaGator 3 is able to provide a PWM signal to both the ESC for the propeller motor and the driver for the stepper motor.

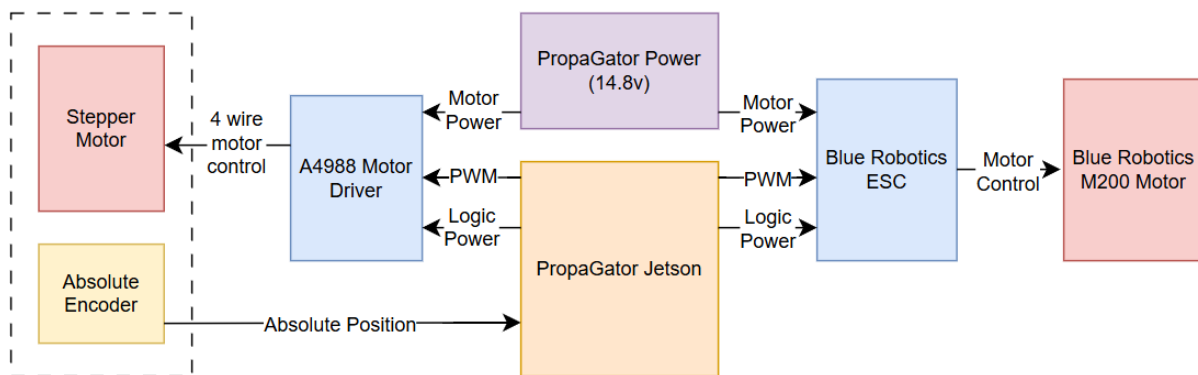


Fig. 2-6. Electrical block diagram

CHAPTER 3

CONTROL SYSTEM

The addition of two azimuth thrusters creates an under-constrained system. With this addition, there are a total of six variables – four thrust magnitudes and two azimuth angles to create a desired velocity state. The thruster allocation equation is given by (7), where w is the width between the fixed thrusters, r is the forward/backward position of the azimuth thrusters, and ϕ_3 and ϕ_4 are the angles of the azimuth thrusters. The desired x , y , and rotational velocity are given by v_x , v_y , and $\dot{\theta}$, respectively.

$$\begin{bmatrix} v_x \\ v_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cos(\phi_3) & \cos(\phi_4) \\ 0 & 0 & \sin(\phi_3) & \sin(\phi_4) \\ -w & w & r \sin(\phi_3) & -r \sin(\phi_4) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \quad (7)$$

Minimum Norm Solution

All six variables must be solved simultaneously for a given velocity state. This can be accomplished by linearizing the system with (8), so the system can be rewritten as (9).

$$a_3 = v_3 \cos(\phi_3) \quad a_4 = v_4 \cos(\phi_4) \quad (8)$$

$$b_3 = v_3 \sin(\phi_3) \quad b_4 = v_4 \sin(\phi_4)$$

$$\begin{bmatrix} v_x \\ v_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ -w & w & 0 & r & 0 & -r \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ a_3 \\ a_4 \\ b_3 \\ b_4 \end{bmatrix} = [A]x \quad (9)$$

A valid solution x_0 can be found simply by using a Moore-Penrose pseudoinverse (10).

$$x_0 = [A]^+ \begin{bmatrix} v_x \\ v_y \\ \dot{\theta} \end{bmatrix} \quad (10)$$

Despite (10) being a valid solution that will result in the desired motion, it does not use any physical meaning and may be non-optimal. This solution gives the minimum norm solution, but as units with different physical meaning (such as linear displacement and angular displacement) are mixed in the $[A]$ matrix, a minimum norm solution may not be equivalent to a minimum effort solution.

Instead, it is desired to find an optimal solution subject to other parameters, in other words to find an optimal set of thrusts and angles, x_{opt} , that still satisfies (9). We can define x_{opt} in (11), where $[N]$ is the basis of the null space of the $[A]$ matrix, and α is a correction vector. The dimension of α is the same as the dimension of x .

$$x_{opt} = x_0 + [N]\alpha \quad (11)$$

$[N]$ is a matrix where each column is a basis vector of the null space of $[A]$. The null space is a set of vectors v such that for all v in the null space, $[A]v = 0$. This is a useful property for modifying the initial solution x_0 to obtain x_{opt} because, as shown in (12) by substituting (11) into (9), when multiplied by the $[A]$ matrix, the resulting solution reduces to the minimum norm solution obtained in (10). The practical interpretation of this is that by choosing a correction vector α , we are modifying how the thrust is distributed while also maintaining the same resultant thrust. Note that in (12) the RHS is identical to the solution obtained in (10).

$$[A]x_{opt} = [A](x_0 + [N]\alpha) = [A]x_0 + [A][N]\alpha = [A]x_0 \quad (12)$$

The problem now becomes finding the correction vector α . Because the $[A]$ equation contains nonlinear terms that all affect each other, an analytical solution is difficult to compute. Instead, a cost function method was chosen to be used.

Cost Function Design

Cost functions are a method of minimizing some value (the cost) by computing the value with different parameters until a minimum is found. There are multiple algorithms to find this minimum. Cost functions also allow a high level of customization, as the cost can be a function of any combination of observable parameters or results.

For this project, the parameters chosen to be minimized are the overall thruster effort, with extra emphasis on minimizing the effort of the azimuth thrusters. The azimuth thrusters have more moving parts that are susceptible to wear, so they should be used only when necessary.

The resulting cost C (13) simply computes the norm of the allocation vector, then adds the components relating to the azimuth thrusters.

$$C = \|x_0 + [N]\alpha\|^2 + a_3^2 + a_4^2 + b_3^2 + b_4^2 \quad (13)$$

The task of finding the minimum cost is then simply a task of finding the α that minimizes C , written in (14).

$$\alpha_{opt} = \arg \min(C(x, \alpha)) \quad (14)$$

For this project, the cost function was implemented in MATLAB. To test the effectiveness of the cost function thruster allocation method, 100 random vectors of five types of thrust request were generated. The minimum norm solution given by (10) and

the corrected solution given by (11) were computed for each test vector. The resulting reduction in azimuth thrust for each test case is given in Table 3-1

Table 3-1. The reduction in azimuth thrust for five different types of thrust request. The 0 reduction for the pure sideways translation type is expected as only the azimuth thrusters can contribute to sideways motion, so the minimum norm and corrected solutions will be equal.

Requested Thrust Type	Azimuth Thruster 3 Thrust Reduction (%)	Azimuth Thruster 4 Thrust Reduction (%)
Pure Translation (Forward/Backward)	33.3	33.4
Pure Translation (Sideways)	0	0
Pure Rotation	23.3	23.3
Combined Translation (Forward/Backward + Sideways)	23.5	23.5
Combined Movement (Combined Translation + Rotation)	23.5	25.1

CHAPTER 4

CONCLUSION AND FUTURE WORK

This project documented the mechanical design of a prototype azimuth thruster for use on the PropaGator 3 Autonomous Surface Vehicle. The prototype addressed and solved limitations of the previous version of the thruster. This new thruster adds the capability to remove the entire propeller assembly for maintenance and repair off the boat. Additionally, the new design greatly reduced the number of unique and fasteners. In the azimuth assembly, the part count was reduced from 29 to 11, and the fastener count was reduced from 36 to 16.

A working prototype was also created to demonstrate the mechanical capabilities of the new version of the thruster. Both the thruster propulsion and azimuth rotation were demonstrated with a benchtop electronics setup.

Future Work

To prepare for the thruster to be deployed on PropaGator 3 at RobotX 2026, there are a small number of modifications that must be made to the thruster and the Blue Boat. A waterproof enclosure must be made for the stepper motor and encoder on top of the thruster. The M200 motor is waterproof out-of-the-box, but the stepper and encoder are not. To seal them, the easiest solution would likely be to add a cover over them with an O-ring. In addition, the interface where the cable for the M200 motor passes through the azimuth assembly must be waterproofed. This could likely be accomplished via a lip seal around the interface.

To allow the propeller assembly and azimuth assembly to be completely detached, a mate-able connector would need to be added to the M200 cable between the slip ring and the motor.

The final possible change would be to machine some components from metal. While most components would not be under a significant load, the propeller ring would likely have a moment induced on it due to drag from the water. In addition to the propeller ring, the azimuth gear would likely benefit from being machined to prolong its life. The propeller itself would likely benefit from being machined from Delrin or a similar material, as was done for PropaGator 2 [1].

After these mechanical changes, further work will need to be determined based on the results of in-water testing.

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https://robonation.org/app/uploads/sites/3/2019/10/UFlorida_RB14_Paper.pdf

BIOGRAPHICAL SKETCH

Adam McAleer was an undergraduate student in mechanical engineering at the University of Florida. Throughout his time at UF, Adam worked in the Machine Intelligence Laboratory on the design, testing, and deployment of autonomous underwater, maritime, and aerial vehicles. Adam served the lab as a mechanical group leader from 2023-2026 and also served as team leader from 2025-2026. His primary technical contributions were to the mechanical design of SubjuGator 9, NaviGator, and PropaGator 3. After graduating from UF in 2026, Adam pursued a master's degree in Robotics at Northeastern University.